INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I , Fourth Semester Examination, February- 2025

Subject Code: BCACC402

Subject Name: Programming with Java

Full Marks: 30 Set No: 1 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2×5)

1. What is Java, and why is it platform-independent? **2**

2. Explain the difference between JDK, JRE, and JVM. **2**

3.. What is the purpose of the static keyword in Java? **2**

4. Explain byte code. **2**

5. What is new in java to create object? **2**

6. Write the output of the following program **2**

```
import java.util.Scanner;
public class SumOfNumbers5
{
public static void main(String args[])
{
int x, y, sum;
Scanner sc = new Scanner(System.in);
System.out.print("Enter the first number: ");
x = sc.nextInt();
System.out.print("Enter the second number: ");
 y = sc.nextInt();
sum = sum(x, y);
System.out.println("The sum of three numbers x, y, and z is: " + sum);
}
 public static int sum(int a, int b)
{
int sum = a + b ; return sum;
}
}
```

165

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

6. Explain why OOP is better compare to procedural programming languages. **5**

7. Write a Java program that takes a year from user and print whether that year is a leap year or not **5**

8. Write a java program to take three numbers from the user and print the greatest number **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Write a program to print the area of a rectangle by creating a class named 'Area' taking the values of its length and breadth as parameters of its constructor and having a method named 'returnArea' which returns the area of the rectangle **10**

11. Write a java program which initialization earning of an employee. The program should calculate the income tax to be paid by the employee as per the criteria given below: **10**
Slab rate IT rate
Up to Rs. 50,000 Nil
up to Rs. 60,000 10% on additional amount up to Rs. 1,50,000 20%
on additional amount Above Rs. 1,50,000 30% on additional amount Hint:
- Run: - java calculates 1,25,000
Result: - income tax is .....

--End--

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Subject Name: Programming with Java

Full Marks: 30 Set No: 2 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1.	What is Inheritance in Java?	2
2	What is super class and subclass?	2
3.	Can we inherit private members of base class to its subclass?	2
4	Can a final method be overridden?	2
5.	Differentiate between static and non-static methods in Java	2

6.	Write the output of the following program
	byte x;
	int a = 270;
	double b = 128.128; x = (byte) a;
	System.out.println("a and x " + a + " " + x);.
		2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7.	Explain the concept of garbage collection in JAVA	5
8.	"Constructor in JAVA can be either public or default"- Justify	5
9.	Design a JAVA based application to display Fibonacci series using two classes	5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10.	Design an application by creating a class named 'AddDistance'. There will be inch and feet as private data members. Add two objects of AddDistance class to add two distances. If inch value is more than 12, 1 will be added with feet.	10
11.	Write down a program to calculate the area of square	10

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and circle using method overloading

--End--

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Subject Name: Programming with Java

Full Marks: 30 Set No: **3** Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1.	What is new in java to create object?	**2**
2.	What is a constructor	**2**
3.	What is Java, and why is it platform-independent?	**2**
4.	Write down the expression of for loop in JAVA.	**2**
5.	What is command line argument?	**2**
6.	What is method overriding?	**2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7.	Write a Java program to take the user for a distance (in meters) and the time taken (as three numbers: hours, minutes, seconds), and display the speed, in meters per second, kilometers per hour and miles per hour (hint: 1 mile = 1609 meters)	**5**
8.	How does Multiple inheritance implement in Java?	**5**
9.	Discuss different types of inheritance is possible in JAVA	**5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10.	Develop two different classes to perform withdraw, deposit and balance check function. Create a main class to call functions to perform operations. Give user the facility to choose the operation	**10**
11	Write down a program to calculate the area of square and circle using method overloading .	**10**